

A proposal for Photonic IC based Mach-Zehnder Interferometer

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Index Terms—PIC(Photonic IC), MZI(Mach-Zehnder Interferometer)

I. INTRODUCTION

THIS report provides the details of Mach-Zehnder Interferometer(MZI) using silicon photonics. Silicon photonics is an emerging field which uses well established fabrication process line for photonics application. Discrete components like fiber to chip coupler, wavelength multiplexers, demodulator, phase change detection etc. can now be integrated into a single photonics chip. This integration of components opens up wide variety of applications ranging from sensing to quantum Many foundaries provide PDK's for standard photonics components and frequent fabrication cycles enabling faster product to market. Many paid and free software tools are available for component and layout design.

II. MACH-ZEHNDER INTERFEROMETER

MZI is used to measure the relative phase difference between two beams which are generated from a single source. MZI can be used for various applications including Fiber Bragg Grating sensing, Inertial sensing etc. This report includes design details, layout design and experimental results of various MZI having different parameters.

III. SOFTWARE TOOLS

This work uses following software tools for MZI design and fabrication.

- Lumerical MODE: For component design and simulation
- Lumerical INTERCONNECT: For photonic circuit design and simulation
- Klayout: For PIC layout design

IV. MZI DESIGN

The main components of MZI are as described in this section including design parameters.

A. Waveguide

This is the main part of any photonic circuits. The waveguide used in this project have following parameters. The waveguide mode simulation is shown in Figure 1. The effective index vs wavelength is shown in Figure 2. The group index vs wavelength is shown in Figure 3.

B. Waveguide compact model

The compact model for waveguide is,

$$n_{eff} = 2.4436 - 1.1292(\lambda - 1550) - 0.0395(\lambda - 1550)^2 \quad (1)$$

TABLE I: Waveguide parameters

Sl. No.	Parameter	Value
1.	Height	220nm
2.	Width	500nm
3.	Polarisation	TE
4.	Mode Support	Single Mode

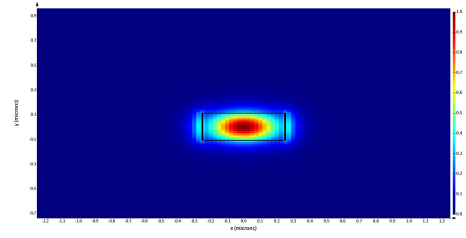


Fig. 1: Waveguide TE Mode simulation

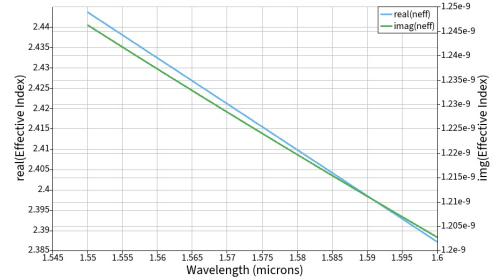


Fig. 2: Effective index vs Wavelength

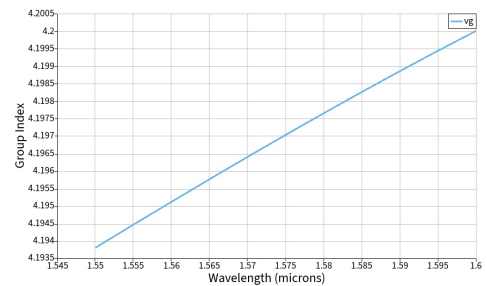


Fig. 3: Group index vs Wavelength

C. MZI design name with parameters

Multiple MZI designs are present in the layout with following details. In addition to MZI, loop waveguides are placed. Two different designs of MZI are also placed using y-branch and directional couplers. The MZI schematics and their corresponding transmission curves are given below.

TABLE II: MZI parameters

Sl. No.	MZI design name	Path difference
1.	MZI1	57.754
2.	MZI2	62.034
3.	MZI3	57.754

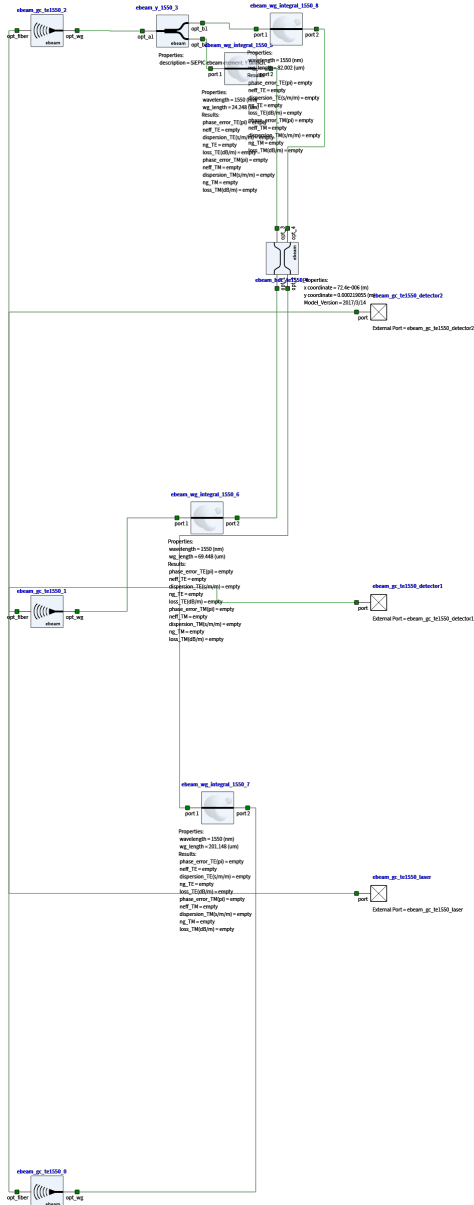


Fig. 4: MZI1 schematic

V. EXPERIMENTAL RESULTS

To be added later

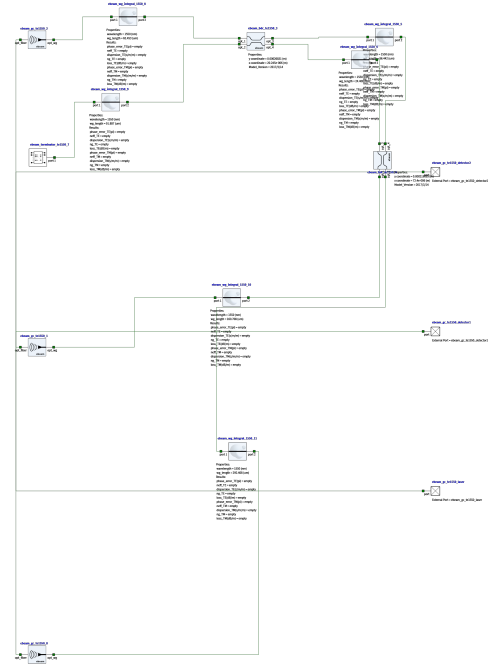


Fig. 5: MZI2 schematic

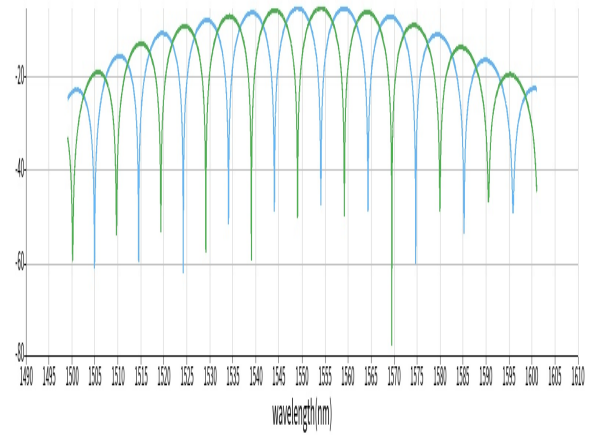


Fig. 6: MZI1 response curve

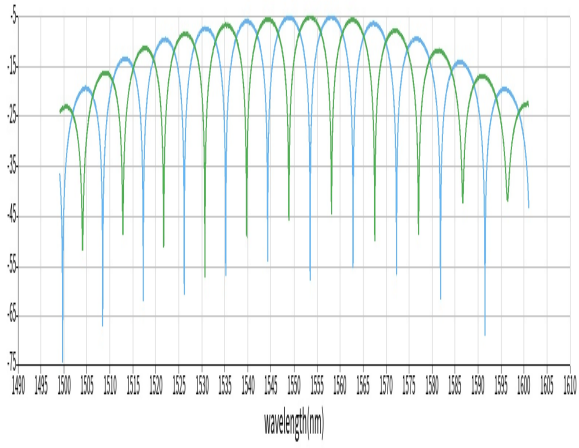


Fig. 7: MZI2 response curve

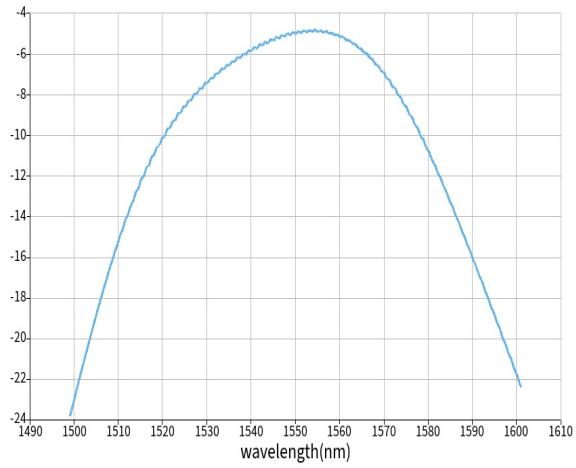


Fig. 8: Loop waveguide response curve

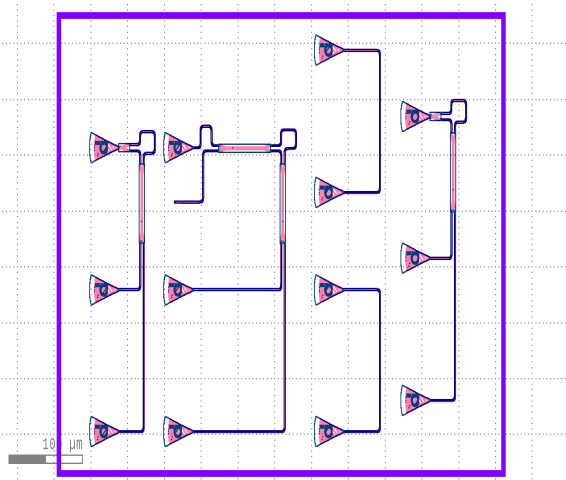


Fig. 9: PIC Layout